Michael Bishop - MAT422

Introduction:

I want to find relationships between different data points in the heart dataset and the heart disease diagnosis. My objective is to create a model for predicting heart disease based on the features in my selected dataset. While my paper in particular may have little significance in the field of heart health, creating models for predicting heart disease in patients can be a great utility for the scientific community allowing doctors to help point patients either to help they may need in the future or to steer a patient away from habits that increase risk of heart disease.

Related Work:

It is highly likely the relationship between heart disease and the features in the heart disease dataset have already been studied. As I complete more of the final project I will look into existing studies on heart disease and compare my analysis to theirs. I have no intent of finding anything ground breaking with this project as I am not informed enough on how the heart operates and would rather see if the results in other studies are repeatable and if I am capable of using the methods I've learned in this course effectively.

To find related works I will use both the ASU library and Google Scholar searching specifically for research on heart health prediction from the past decade, if the research for some reason uses the same dataset I used I will attempt to avoid it so as not to directly copy their methodology, though that is unlikely to need to occur.

Proposed Methodology:

There exist several methods to compare the relationship between variables in data science. I will check the correlation of each feature with the heart disease target variable to see how correlated the variables are, and will use linear regression to attempt to predict what heart disease value an individual may have based on the feature variables. It may also be good to classify individuals based on having or not having heart disease and then using logistic regression to get a probability for each individual that the person has heart disease.

As I complete the remainder of this course it is not out of the question that I learn about new methods that will also be useful for finding relationships between the features and heart disease in the dataset and predicting heart disease based on the features of the dataset.

Once I use these techniques to create functions that predict heart disease I can use metrics like RMSE for linear regression, or a confusion matrix for logistic regression to see how often I can accurately predict heart disease in the sample set.

Experiment Setup:

I will use the heart disease dataset and filter out any null values when necessary. This dataset includes categorical, continuous (real number), and integer data. I will use python to filter and analyze the data, as well as draw graphs and diagrams for my paper. Using the methods from the Proposed Methodology section I can create the aforementioned graphs to represent how the features in the dataset relate to heart disease.

Expected Results:

I anticipate that my findings will match closely with existing literature on the topic. Additionally I anticipate that variables such as age and cholesterol will impact the likelihood that a patient will have heart disease. It would not surprise me if sex had little impact on heart disease, but without analyzing the data I'm not sure. I expect to be able to predict with a decent degree of accuracy (>80%) whether a patient has heart disease.

Team Roles:

Since I am working alone I will be responsible for every part of the project from filtering data to finding data to presenting findings.